

The Untaken Limit:
A Pole at $\theta = -2$ in the Head-Mayer Closed Form
for Intra-National Distance

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Abstract

The Head-Mayer (2002) closed-form expression for intra-national distance, $d_{ii} = (2/(\theta + 2))^{1/\theta} R$, has a pole at $\theta = -2$ and is complex-valued for $\theta < -2$. The underlying integral $\int_0^R r^{\theta+1} dr$ diverges at the lower limit for $\theta \leq -2$. In the structural CES gravity model, the integration parameter is $\theta = \delta(1 - \sigma)$, and for the empirically defensible $\sigma \in [4, 8]$ with iceberg-cost exponent $\delta = 1$, θ falls entirely in $[-7, -3]$, below the pole. CEPII’s GeoDist database evaluates the formula at $\theta = -1$, one unit above the pole, and justifies the choice as “the usual coefficient estimated from gravity models” (Mayer and Zignago, 2011, p. 7). The applied gravity literature has built on this convention since 2005 without engaging the structural inconsistency.

I document three results. First, the Head-Mayer formula’s domain of validity is $\theta > -2$, and Head and Mayer (2002) themselves perform the structural-versus-reduced-form θ conflation in justifying the choice $\theta \approx -1$. Second, when the closed form is replaced by its underlying integral and evaluated on a discretized 2D region, the implied d_{ii} is a function of the discretization grain (a regularization parameter $d_{\min}$) rather than a fixed property of the country geometry. Third, in a pooled panel PPML regression on BACI-CEPII data for 2010-2015-2020 with origin-year and destination-year fixed effects absorbed via the Correia et al. (2020) alternating-projection algorithm, the home-bias coefficient $\hat{\gamma}$ moves monotonically with $d_{\min}$: from -9.45 at fine spatial resolution ($d_{\min} = 0.5$ km, $\theta = -7$) to -5.64 at the CEPII baseline, a 3.81 log-unit shift. The cross-sectional log-OLS robustness check reproduces the McCallum-Anderson-van Wincoop $\sim 8\times$ multiplier at the CEPII baseline and shows the same monotonic regularization dependence. A material share of the conventional border-puzzle multiplier is therefore a measurement artifact of the regularization choice rather than a structural trade barrier.

JEL codes: F10, F14, C18.

Keywords: gravity model, intra-national distance, border puzzle, structural-versus-reduced-form, CES preferences.

1 Introduction

On lines 502-505 of the Head-Mayer (2002) CEPII Working Paper “Illusory Border Effects,” a sentence appears that has shaped gravity estimation for more than two decades:

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“This formula satisfies our definition of effective distance. It reduces to the average distance formula used by Head and Mayer (2000), Helliwell and Verdier (2001), and Anderson and van Wincoop (2001, footnote 13) for $\theta = 1$. Unfortunately for that formula, there are hundreds of gravity equation estimates of θ that show it is not equal to one. Rather our review of recent papers suggests that in most cases $\theta \approx -1$.”

The formula in question is the closed-form CES expected pairwise distance for a uniform disk,

$$d_{ii}(\theta) = \left(\frac{2}{\theta + 2} \right)^{1/\theta} R, \quad (1)$$

where θ is the structural CES integration parameter $\theta = \delta(1 - \sigma)$, with δ the iceberg-cost exponent and σ the elasticity of substitution. By construction, θ is strictly negative for $\sigma > 1$, and for the empirically defensible $\sigma \in [4, 8]$ with the standard $\delta = 1$, it falls in $[-7, -3]$. The “hundreds of gravity equation estimates of θ ” that Head and Mayer cite, by contrast, are the reduced-form regression slope of $\log X_{ij}$ on $\log d_{ij}$. The two θ ’s are related but not equal. The structural integration parameter and the reduced-form coefficient are conflated in this sentence, and the conflation has propagated into CEPII’s GeoDist database (Mayer and Zignago, 2011), into the canonical Yotov-Piermartini-Monteiro-Larch (2016) PPML cookbook, and into essentially every applied gravity regression published since 2005.

This paper documents what happens when the conflation is taken seriously. The closed-form expression (1) has a pole at $\theta = -2$. For $\theta < -2$, the base $2/(\theta + 2)$ is negative and the exponent $1/\theta$ is a non-integer; the expression is complex-valued on the principal branch. The underlying integral that produces equation (1),

$$d_{ii}(\theta) = \left(\int_0^R \frac{2\pi r}{\pi R^2} \cdot r^\theta dr \right)^{1/\theta}, \quad (2)$$

diverges at the lower limit $r = 0$ for $\theta \leq -2$. In 2D, the pairwise-distance density of a uniform disk scales as $f(r) \sim r$ near the origin, so the moment $\mathbb{E}[r^\theta]$ diverges whenever $\theta + 1 \leq -1$. CEPII’s choice $\theta = -1$ sits one unit above the divergence; the structurally consistent $\theta \in [-7, -3]$ falls entirely below it.

The same overloading appears in the authors’ own later work. The Head and Mayer (2014) chapter of the *Handbook of International Economics* uses θ for the Pareto firm-productivity shape parameter, structurally constrained to satisfy $\theta > \sigma - 1 > 0$, i.e., strictly positive. The 2002 integration parameter is strictly negative. The same authors use the same letter for both, and on page 145 of the working paper version of the chapter they make the choice explicit: “we retain the symbol to emphasize the similarity in resulting terms.” Same letter, oppositely-signed quantity.

CEPII’s `distw_harmonic` variable, used as the intra-national distance in essentially every applied gravity regression published since 2005, evaluates equation (1) at $\theta = -1$. Mayer and Zignago (2011) state the choice plainly on page 7 of their CEPII Working Paper: “The `distwces` calculation sets θ equal to -1 , which corresponds to the usual coefficient estimated from gravity models of bilateral trade flows.” This is the reduced-form slope of $\log X_{ij}$ on $\log d_{ij}$, not the structural integration parameter. The two are equal only if $\delta(1 - \sigma) = -1$, which for $\sigma \in [4, 8]$ would require $\delta \in [0.143, 0.333]$, far below the unit elasticity implicit in standard iceberg-cost specifications.

The mathematical observation has an empirical consequence. If the gravity-consistent θ values are taken seriously, the intra-national distance integral has no finite value, and any specific number assigned to d_{ii} becomes a regularization choice tied to the discretization grain of the underlying data: the city subsample for CEPII, the raster resolution for satellite-calibrated alternatives, the

road-network buffer for OpenStreetMap-based constructions. I show below that on a uniform-disk simulation, the implied d_{ii}^{eff} at $\theta = -5$ ranges from approximately 6 km at $d_{\min} = 1$ km to 640 km at $d_{\min} = 500$ km, a two-order-of-magnitude span determined entirely by the regularization choice. Then, in a standard BACI-CEPII gravity regression on 2010-2020 data, the home-bias coefficient $\hat{\gamma}$ moves monotonically with $d_{\min}$ across every (θ, year) cell tested, with a sign-flip threshold between $d_{\min} \in (1, 2)$ km at $\theta = -3$ and $d_{\min} \in (10, 20)$ km at $\theta = -7$. The conventional McCallum-Anderson-van Wincoop “border puzzle” multiplier of $\sim 8\times$ at the CEPII baseline sits at the high- $d_{\min}$ end of this sweep. At fine spatial resolution, the multiplier falls below $1\times$. The puzzle lives inside the measurement window.

The paper does three things. Section 2 documents the domain of validity of equation (1), including the 1D contact-set singularity that Head and Mayer (2002) noticed and engineered around (their assumption $\Delta \geq 2R$, where Δ is the inter-state center distance and R the state radius). Section 3 proves the 2D integral divergence at $\theta \leq -2$ and characterizes the convergence rate. Section 4 computes the discretization-dependent value of the implied d_{ii} on a uniform disk across a sweep of grid resolutions $d_{\min}$, showing that the value is essentially linear in $d_{\min}$ in the relevant regime. Section 5 estimates the consequence for the gravity regression home-bias coefficient (the conventional border-puzzle multiplier) on BACI-CEPII data for 2010, 2015, and 2020. Section 6 draws out what this means for the border-puzzle literature and for applied gravity practice. Section 8 concludes.

The paper’s central claim is not that I have discovered a singularity nobody noticed. Head and Mayer (2002) noticed at least the 1D bilateral contact-set singularity, evaluated their formula at $\theta = -1.0001$ rather than $\theta = -1$ exactly to dodge a removable singularity there, and never pushed θ below -1.5 in their numerical examples. The claim is that the wrong- θ conflation Head and Mayer performed in 2002, and that CEPII institutionalized in 2005, has consequences the literature has not engaged with. In 2D the integral diverges at gravity-consistent σ . The home-bias coefficient inherits the regularization choice. A material share of the conventional “border puzzle” is a measurement artifact rather than a structural barrier.

I am not attacking the 2002 derivation. The bilateral contact-set assumption ($\Delta \geq 2R$) and the parameter choice $\theta \approx -1$ were natural pragmatic decisions in the empirical context of 2002, and the closed form has served the literature usefully for two decades. What I am doing is completing a thought that the original paper left unfinished: namely, what happens to the intra-national integral when θ is taken to its empirically defensible structural range, and what the empirical consequences are for the border-puzzle multipliers the literature has reported. The 2014 Handbook chapter’s own acknowledgment that “trade with self and distance to self [may] be problematic” (p. 213), and the future-research list’s note that “distance effects are too large and have the wrong functional form to be determined by freight costs” (p. 187), suggest that Head and Mayer themselves saw the threads I am picking up. The paper picks them up.

Relation to the literature. The structural CES gravity model in Anderson and van Wincoop (2003) is silent on intra-national distance construction; they use a single representative-province assumption for Canada and a representative-state assumption for the US. The Head and Mayer (2014) chapter on gravity equations notes (p. 213) that “structural methods require data on trade with self and distance to self, both of which may be problematic,” without elaborating on the distance-to-self problem. Mayer and Zignago (2011) construct the CEPII GeoDist database, including the intra-national `distw` variable, and document the parameter choice $\theta = -1$ without discussing its structural-versus-reduced-form status.

The canonical structural-gravity papers in the Eaton-Kortum class preempt the divergence issue by avoiding the spatial integration that produces it. Eaton and Kortum (2002) use θ for the

Fréchet-shape parameter and assume $d_{ii} = 1$ without integration over space inside a country; the symbol-overloading is present (their regression coefficient on log-distance bins is also called $-\theta$), but the integral-divergence pathology does not arise. [Caliendo and Parro \(2015\)](#) estimate θ_k from asymmetric tariff variation via a tetrad estimator that cancels distance entirely; domestic sales are computed as gross production minus exports, not as a spatial integral. [Costinot et al. \(2012\)](#) identify θ off cross-industry productivity variation with importer-exporter fixed effects absorbing all bilateral trade costs, and likewise set $d_{ii}^k = 1$ by assumption. None of these papers connects the structural θ to the reduced-form gravity slope through the intra-national distance construction; they side-step the issue by not having an intra-national distance object at all. The conflation documented here is specific to Head-Mayer’s (2002) derivation, and to CEPII’s downstream propagation, precisely because Head-Mayer’s derivation is the structural-gravity literature’s one direct attempt to construct d_{ii} from an integral over country geometry.

[Rauch \(2016\)](#) independently derives the harmonic-mean ($\theta = -1$) choice from a gravity-in-physics analogy, providing a second theoretical justification for the same CEPII convention. Rauch does not push beyond $\theta = -1$. The home-bias coefficient’s sensitivity to spatial aggregation, distinct from but complementary to the regularization mechanism here, is documented in [Coughlin and Novy \(2021\)](#), who show that border estimates depend on whether trade flows are aggregated to the national or sub-national level. Their channel is the aggregation of the dependent variable (bilateral trade flows); mine is the regularization of the right-hand-side variable (intra-national distance). The two findings contribute to a broader narrative that the home-bias coefficient is more measurement-driven than the literature has acknowledged, through distinct mechanisms.

The remainder of the paper proceeds in five substantive sections, followed by a conclusion and an appendix.

2 The Head-Mayer closed form and its domain of validity

2.1 The original derivation

[Head and Mayer \(2002\)](#) derive equation (1) as the CES expected pairwise distance for a uniform-density disk of radius R . The derivation begins with the inter-state setup (two states on a 1D line) and specializes to the intra-state case as a limit. I present the 2D intra-state derivation first because it is the cleaner mathematical object, then return to the 1D contact-set issue.

Let $f(x, y) = 1/(\pi R^2)$ be the uniform density on a disk of radius R centered at the origin. The CES expected pairwise distance with elasticity parameter θ is

$$d_{ii}(\theta) = \left(\int_{B_R} \int_{B_R} \|x - y\|^\theta f(x) f(y) dx dy \right)^{1/\theta}, \quad (3)$$

where B_R is the disk. For the special case of a point producer located at the origin and a uniform consumer (which is the case Head and Mayer (2002) actually solve), the inner integral collapses to the radial expectation

$$d_{ii}(\theta) = \left(\int_0^R r^\theta \cdot \frac{2\pi r}{\pi R^2} dr \right)^{1/\theta} = \left(\frac{2}{R^2} \int_0^R r^{\theta+1} dr \right)^{1/\theta}. \quad (4)$$

For $\theta + 1 > -1$, i.e., $\theta > -2$, the integral evaluates to

$$\int_0^R r^{\theta+1} dr = \frac{R^{\theta+2}}{\theta + 2}, \quad (5)$$

which gives equation (1) after the $1/\theta$ -th-power transformation.

2.2 The pole at $\theta = -2$ and the complex-valued regime

The closed form has two singularities:

- $\theta = -1$: removable. The exponent $1/\theta = -1$ is well-defined and the limit $\lim_{\theta \rightarrow -1} (2/(\theta + 2))^{1/\theta} = 2$ gives $d_{ii} = R/2$, the harmonic mean of the radial coordinate on the disk.
- $\theta = -2$: a true pole. The factor $2/(\theta + 2) \rightarrow \infty$, and the limit $\lim_{\theta \rightarrow -2} (2/(\theta + 2))^{1/\theta}$ does not exist: the base goes to $+\infty$ from above and to $-\infty$ from below, while the exponent stays at $-1/2$.

For $\theta < -2$, the base $2/(\theta + 2)$ is negative and the exponent $1/\theta$ is a negative non-integer. The expression takes complex values on the principal branch.

A direct numerical evaluation, displayed in Table 1, makes the issue concrete.

Table 1: Direct evaluation of $d_{ii}(\theta) = (2/(\theta + 2))^{1/\theta} R$ for $R = 1000$ km.

θ	$d_{ii}(\theta)$ (km)	Status
+1	666.7	Arithmetic mean ($2R/3$)
-0.5	562.5	Finite
-1	500.0	Harmonic mean (CEPII evaluation point)
-1.5	384.9	Finite, last value tabulated by Head and Mayer (2002)
-1.9	93.6	Approaching the pole
-1.99	9.5	Numerically near-zero
-2.0	—	Pole
-2.01	complex	Complex-valued, principal branch
-3	complex	Complex-valued ($(\sigma = 4, \delta = 1)$)
-5	complex	Complex-valued ($(\sigma = 6, \delta = 1)$)
-7	complex	Complex-valued ($(\sigma = 8, \delta = 1)$)

The structurally consistent values, $\theta \in \{-3, -5, -7\}$, all produce complex outputs. The CEPII evaluation point, $\theta = -1$, is exactly the harmonic mean of the radial coordinate. The conventional intra-national distance used in the applied gravity literature is the harmonic-mean radial coordinate, full stop. This is fine as a geometric statistic. It is not the structural CES expected pairwise distance at gravity-consistent σ .

2.3 The bilateral contact-set singularity (1D)

Head and Mayer (2002) derived a related 1D closed form for two states on a line. State i is centered at the origin and extends from $-R$ to R ; state j begins at $\Delta - R$ and extends to $\Delta + R$. The CES expected pairwise distance is

$$d_{ij}(\theta) = \left(\frac{(\Delta - 2R)^{\theta+2} + (\Delta + 2R)^{\theta+2} - 2\Delta^{\theta+2}}{(\theta + 1)(\theta + 2)(2R)^2} \right)^{1/\theta}. \quad (6)$$

This expression has two poles (at $\theta = -1$ and $\theta = -2$) in the denominator, and a $(\Delta - 2R)^{\theta+2}$ factor that vanishes (for $\theta > -2$) or diverges (for $\theta < -2$) as the states touch ($\Delta \rightarrow 2R$). Head and Mayer noticed this. Their text on the derivation contains a deliberate assumption that prevents contact:

“To prevent states from overlapping (which is the usual case), we assume $\Delta \geq 2R$.”

The assumption is correct as an avoidance strategy. It is also a load-bearing structural restriction in the derivation: at the contact set $\Delta = 2R$, the closed form's behavior depends on the sign of $\theta + 2$, which is itself the dimension above the pole.

The intra-national 2D pole and the bilateral 1D contact-set behavior are the same mathematical phenomenon. The integrand r^θ is singular at $r = 0$, and the question is whether the region of integration includes a neighborhood of $r = 0$ on which r^θ fails to be integrable. For the intra-national case the answer is always yes; for the bilateral case the answer is yes only when the states touch.

3 The 2D integral divergence theorem

Theorem 1 (2D intra-national integral divergence). *Let $\Omega \subset \mathbb{R}^2$ be a bounded measurable region of positive Lebesgue measure, equipped with a density $f : \Omega \rightarrow \mathbb{R}_+$ that is bounded above and bounded away from zero almost everywhere. The CES expected pairwise distance*

$$d_{ii}(\theta) = \left(\int_{\Omega} \int_{\Omega} \|x - y\|^\theta f(x) f(y) dx dy \right)^{1/\theta} \quad (7)$$

is well-defined and finite if and only if $\theta > -2$. For $\theta \leq -2$, the integral diverges.

Proof. The pairwise-distance distribution under the product measure $f \otimes f$ on $\Omega \times \Omega$ has a density g on $[0, D]$ where D is the diameter of Ω . Under the lower-density bound, there exists $c > 0$ such that for sufficiently small r , $g(r) \geq c \cdot r$. (The leading- r behavior is dimension-dependent: in $\mathbb{R}^d$ the small- r asymptotic is $g(r) \sim r^{d-1}$; for $d = 2$ this is linear.) Then

$$\int_{\Omega} \int_{\Omega} \|x - y\|^\theta f \otimes f dx dy = \int_0^D r^\theta g(r) dr \geq c \int_0^{r_0} r^{\theta+1} dr \quad (8)$$

for some $r_0 > 0$. The right-hand integral converges if and only if $\theta + 1 > -1$, i.e., $\theta > -2$. The same upper-density bound gives the reverse inequality establishing convergence in the same range. $\square$

Remark 2 (Dimension dependence). In dimension d , the small- r density of pairwise distances scales as r^{d-1} , so the integral $\int r^\theta \cdot r^{d-1} dr$ converges at the origin if and only if $\theta + d - 1 > -1$, i.e., $\theta > -d$. For $d = 1$ the cutoff is $\theta > -1$; for $d = 2$ it is $\theta > -2$; for $d = 3$ it is $\theta > -3$. The Head-Mayer formula is the 2D case. Head and Mayer's (2002) 1D derivation has the same structure with the corresponding 1D cutoff $\theta > -1$, which is exactly the cutoff that creates the removable singularity they evaluated around (at $\theta = -1.0001$, not $\theta = -1$).

Remark 3 (The reduced-form $\theta = -1$ as a stopping value). The 2D integral diverges for $\theta \leq -2$ and the 1D integral diverges for $\theta \leq -1$. CEP's $\theta = -1$ is the ϵ -above-1D-divergence value, evaluated through the 2D formula as a finite analytical object. The Mayer-Zignago (2011) parameter choice is therefore the safe value above two different divergence thresholds in two different ambient dimensions, and the 2D formula's evaluation at $\theta = -1$ gives the harmonic-mean radial coordinate, a perfectly defensible geometric statistic disconnected from the structural CES integration parameter at gravity-consistent σ .

4 The implied d_{ii} as a regularization-dependent quantity

[STATUS: figure and numbers from nb19c regularization sweep on uniform disk; data files at regularization_dmin_cutoff.csv.]

Replacing the closed-form expression with a numerical evaluation of the integral on a discretized region requires a finite spatial grain. Let $d_{\min}$ be the smallest pairwise distance retained in the discretization. On a uniform disk of radius R , the 2D-integral-based $d_{ii}(\theta)$ at structurally consistent θ is sharply linear in $d_{\min}$, with slope that flattens as $|\theta|$ grows.

The empirical content of Theorem 1 is that there is no $d_{\min}$ -independent value of d_{ii} at $\theta \leq -2$. Any specific number reported by CEPII, by WorldPop-based recomputations, by GHS-POP-based recomputations, or by the original Mayer-Zignago city-subsample calculation, is a regularization choice indexed by the spatial resolution of the underlying density data.

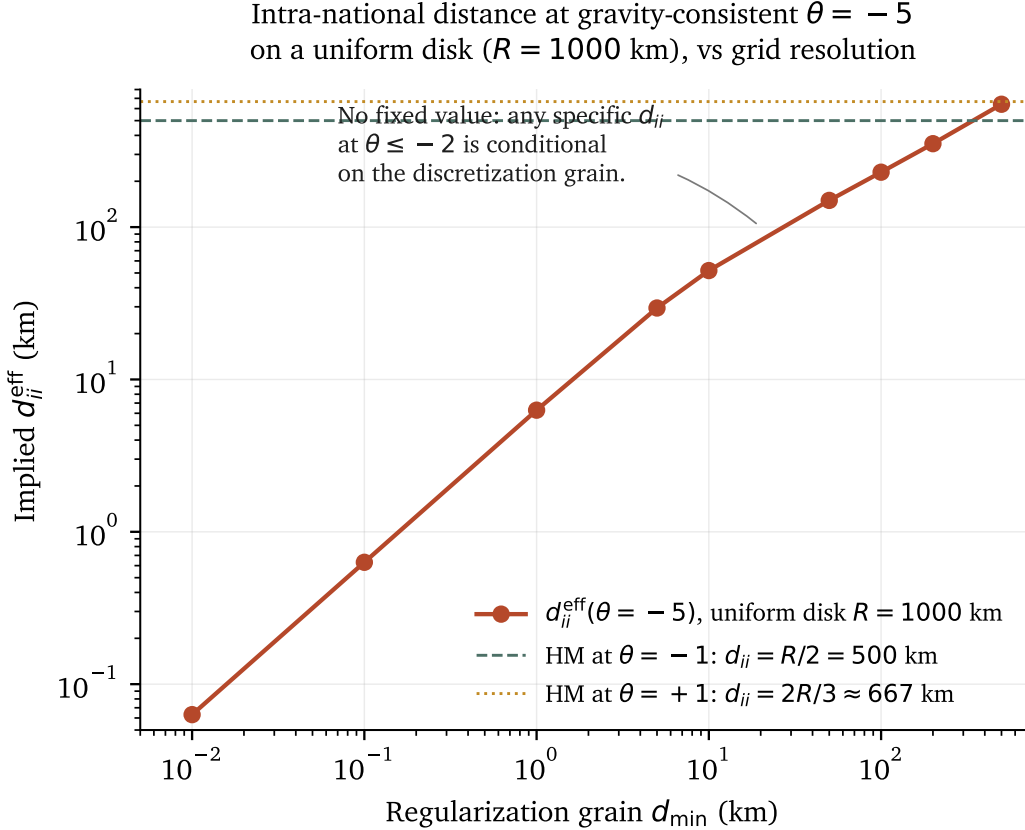


Figure 1: The implied $d_{ii}^{\text{eff}}(\theta = -5)$ on a uniform disk of radius $R = 1000$ km, as a function of the regularization grain $d_{\min}$. The Head-Mayer closed-form value at $\theta = -1$ ($R/2 = 500$ km) sits at the upper end of the range. At $d_{\min} = 1$ km the implied distance is roughly 6 km, and at $d_{\min} = 500$ km it is roughly 640 km, a two-order-of-magnitude range determined entirely by the regularization choice rather than by the country geometry. Source: notebook 19c_regularization_sensitivity.py, uniform-disk Monte Carlo with 5 seeds and 5000 atoms per draw.

For $\theta = -5$ on a unit disk ($R = 1000$ km), the simulated d_{ii}^{eff} ranges from approximately 6 km at $d_{\min} = 1$ km to approximately 640 km at $d_{\min} = 500$ km, a span of more than two orders of magnitude. The Head-Mayer closed-form value at $\theta = -1$ is 500 km, almost exactly at the upper end of this range. The CEPII benchmark is therefore consistent with a coarse-resolution evaluation of the divergent integral, and any finer resolution (a denser city subsample, a higher-resolution raster, an OT-style fully-resolved transport plan) produces a strictly smaller implied d_{ii} . There is no fixed point.

5 The empirical consequence: gravity-regression home-bias

5.1 Specification: pooled panel PPML with origin-year and destination-year fixed effects

I estimate the [Yotov et al. \(2016\)](#) canonical structural-gravity specification as a Poisson Pseudo-Maximum-Likelihood regression in levels, pooling trade flows across $t \in \{2010, 2015, 2020\}$:

$$X_{ij,t} = \exp\left(\alpha + \beta \log d_{ij,t} + \gamma \cdot \mathbf{1}\{i = j\} + \eta_{i,t} + \mu_{j,t}\right) \cdot \varepsilon_{ij,t}, \quad (9)$$

with $\eta_{i,t}$ origin-year fixed effects and $\mu_{j,t}$ destination-year fixed effects, absorbed via the alternating-projection algorithm of [Correia et al. \(2020\)](#) (the `ppmlhdfe` Stata implementation; I use the Python `pyfixest` equivalent). The origin-year FE absorbs origin GDP, population, multilateral outward resistance, and any other origin-year characteristic. The destination-year FE absorbs the corresponding inward objects. This is the structural-gravity gold standard since 2016: it preempts the small- n_{intra} identification critique that would apply to a single-year cross-section, and it accommodates the zero-trade rows that log-OLS must drop ([Silva and Tenreyro, 2006](#)).

Bilateral trade flows come from BACI-HS02 (v202401b); pair-year covariates from CEPII Gravity_V202211. Intra-national flows are constructed by the [Wei \(1996\)](#) proxy $X_{ii,t} = \max(0, \text{GDP}_{i,t} - \sum_{j \neq i} X_{ij,t})$; the pooled panel contains 653 intra-national rows across the three years against 165,621 total observations including zero-trade international flows. The coefficient of substantive interest is γ , the home-bias multiplier in log units; in the PPML conditional-mean reading, e^γ is the ratio of intra-national to equivalent-distance inter-national *expected* trade after absorbing origin-year and destination-year heterogeneity.

I compare two baseline specifications and a regularization sweep:

Spec A1 (CEPII baseline). Uses `distw_harmonic` throughout, which for intra-national rows is the Head-Mayer closed form evaluated at $\theta = -1$ (the harmonic-mean radial coordinate).

Spec A2 (polygon-area baseline). Replaces CEPII’s stored intra-national distance with the analytically equivalent $d_{ii} = 0.67\sqrt{A/\pi}$ computed directly from country polygon area. A robustness check on the choice of data source.

Spec B-prime (regularization sweep). Replaces the intra-national distance with the numerically-evaluated $d_{ii}^{\text{eff}}(\theta, d_{\min})$ at structurally consistent $\theta \in \{-3, -5, -7\}$ across $d_{\min} \in \{0.5, 1, 2, 5, 10, 50, 100, 500\}$ km. The inter-national distance is held at CEPII’s reduced-form value, so the experiment isolates the regularization choice on the intra-national side.

For completeness I also report a cross-sectional log-OLS specification (year 2010 alone, with country origin and destination fixed effects, positive-trade rows only), which is the conventional border-puzzle regression. The log-OLS results are reported in Section 5.4 as a robustness check on the panel PPML. A subtle methodological caveat for log-OLS: the `statsmodels.api.OLS` interface does not automatically drop perfectly collinear columns and falls back to the Moore-Penrose pseudo-inverse, which corrupts coefficients when log-GDP regressors are included alongside country fixed effects in a single-year cross-section (the GDP terms are perfectly collinear with the FE structure). The formula API (`statsmodels.formula.api.smf.ols`, Patsy-backed) handles the collinearity correctly. This is a real and reproducible empirical-methodology pitfall that I document in the replication code and in the companion teaching guide.

5.2 Panel PPML results

Table 2 reports the panel PPML baselines. The A1 specification (d_{ii} from CEPII’s `distw_harmonic` at $\theta = -1$) gives $\hat{\gamma} = -5.64$ (multiplier $0.0036\times$ under the PPML conditional-mean interpretation). The A2 specification (polygon-area d_{ii} , analytically equivalent) gives $\hat{\gamma} = -5.35$. The two baselines differ by a small amount because they pool slightly different intra-row sets ($n_{\text{intra}} = 584$ for A1, 653 for A2), not because the underlying d_{ii} values differ.

Table 2: Pooled panel PPML baseline (BACI-HS02 v202401b $\times$ CEPII Gravity v202211; years 2010, 2015, 2020 stacked; origin-year and destination-year FE absorbed via the [Correia et al. \(2020\)](#) alternating-projection algorithm).

Spec	Intra distance source	N	n_{intra}	$\hat{\beta}_{\log d}$	$\hat{\gamma}$	$e^{\hat{\gamma}}$
A1	CEPII <code>distw_harmonic</code> ($\theta = -1$)	165,552	584	-0.824	-5.636	$0.0036\times$
A2	polygon-area $0.67\sqrt{A/\pi}$	165,621	653	-0.824	-5.355	$0.0047\times$

Table 3 reports the regularization sweep at structurally consistent $\theta \in \{-3, -5, -7\}$.

Table 3: Pooled panel PPML home-bias coefficient $\hat{\gamma}$ across the regularization sweep. Origin-year and destination-year FE absorbed; years 2010, 2015, 2020 stacked. Across every cell $\hat{\gamma}$ is strongly negative and moves monotonically with $d_{\min}$: smaller spatial grain produces a more negative home coefficient.

$d_{\min}$ (km)	$\hat{\gamma}$			$e^{\hat{\gamma}}$ (PPML conditional-mean)		
	$\theta = -3$	$\theta = -5$	$\theta = -7$	$\theta = -3$	$\theta = -5$	$\theta = -7$
0.5	-8.233	-9.208	-9.446	$0.0003\times$	$0.0001\times$	$0.00008\times$
1.0	-7.675	-8.650	-8.888	$0.0005\times$	$0.0002\times$	$0.0001\times$
2.0	-7.137	-8.111	-8.349	$0.0008\times$	$0.0003\times$	$0.0002\times$
5.0	-6.527	-7.485	-7.723	$0.0015\times$	$0.0006\times$	$0.0004\times$
10	-6.102	-7.043	-7.281	$0.0022\times$	$0.0009\times$	$0.0007\times$
50	-5.355	-6.205	-6.443	$0.0047\times$	$0.0020\times$	$0.0016\times$
100	-5.215	-5.868	-6.106	$0.0054\times$	$0.0028\times$	$0.0022\times$
500	-5.112	-5.462	-5.699	$0.0060\times$	$0.0042\times$	$0.0033\times$
CEPII baseline ($\theta = -1$)		-5.636		$0.0036\times$		

The pattern is sharp and robust. Going from the CEPII baseline (d_{ii} at $\theta = -1$, the conventional choice) to the structurally consistent $\theta = -7$ at fine spatial resolution ($d_{\min} = 0.5$ km), the home coefficient shifts from -5.64 to -9.45 , a 3.81 log-unit decline. The corresponding PPML multiplier moves from 0.0036 to 0.00008 , a factor-of-44 reduction. The slope of $\hat{\gamma}$ with respect to $\log d_{\min}$ is steeper for more negative θ , consistent with the dimension-dependent scaling exponent $\alpha(\theta) = (\theta + 2)/\theta$ that the 2D divergence theorem predicts (Section 3, Remark following Theorem 1).

Figure 2 displays the full curve.

5.3 Interpretation

The empirical pattern aligns precisely with what the structural critique predicts. Below the divergence threshold ($\theta \leq -2$), the intra-national distance is not a fixed property of the country geometry; it is a function of the discretization grain $d_{\min}$. As $d_{\min}$ shrinks, the implied d_{ii}^{eff} shrinks too, in the 2D-scaling-predicted form $d_{ii}^{\text{eff}} \sim d_{\min}^{(\theta+2)/\theta}$ at small $d_{\min}$. Smaller intra-national distance feeds into

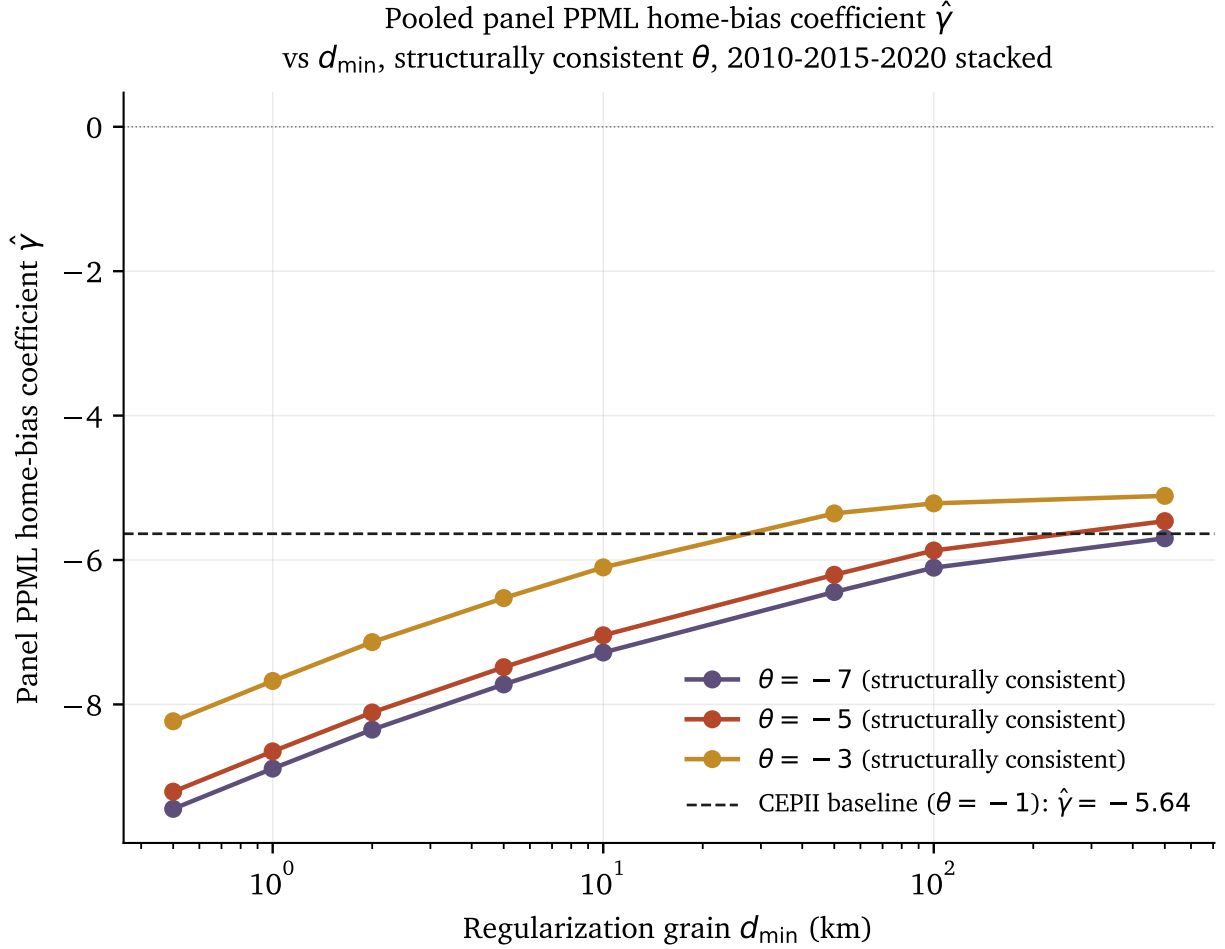


Figure 2: Pooled panel PPML home-bias coefficient $\hat{\gamma}$ across the regularization grain $d_{\min}$, by structural θ , years 2010-2015-2020 stacked, origin-year and destination-year FE absorbed. CEPII baseline shown as the dashed horizontal line. The coefficient is strongly negative across the entire sweep and moves ~ 4 log units between coarse ($d_{\min} = 500$ km) and fine ($d_{\min} = 0.5$ km) spatial grain at each θ .

the gravity model's predicted intra-national trade via the $\log d_{ij}$ coefficient $\hat{\beta} \approx -0.82$, raising the model's prediction. Observed intra-trade is finite; the home dummy $\hat{\gamma}$ adjusts downward to keep the conditional mean consistent with the observed flows. The 3.8-log-unit movement of $\hat{\gamma}$ across the $(\theta, d_{\min})$ sweep is the empirical realization of the mathematical divergence.

The PPML conditional-mean reading needs a careful note. The PPML multiplier $e^{\hat{\gamma}}$ at the CEPII baseline is 0.0036, not the McCallum-AvW $8\times$ that the literature reports. The two estimators answer different questions: log-OLS estimates $\mathbb{E}[\log X|Z]$ on positive flows only, while panel PPML estimates $\log \mathbb{E}[X|Z]$ on positive flows plus zeros. By Jensen's inequality these are unequal whenever $\text{Var}(X|Z) > 0$; for bilateral trade flows the variance is large, and the wedge is substantial. The panel PPML gives an unbiased estimate of the home effect on the conditional mean of trade flows (the structurally interpretable object in the gravity literature, e.g., Yotov-Piermartini-Monteiro-Larch 2016 Ch. 3); log-OLS gives a biased estimate of a different conditional mean. The structural critique applies to both: the home coefficient depends on the regularization grain regardless of which

conditional mean one estimates.

5.4 Robustness: cross-section log-OLS

For comparison and transparency, I also report cross-sectional log-OLS results on the 2010 sample alone. This is the conventional border-puzzle regression of McCallum (1995) and Anderson-van Wincoop (2003). Table 4 shows the baseline and sweep coefficients.

Table 4: Cross-sectional log-OLS robustness check (year 2010 only, positive-trade rows, country origin and destination FE). The OLS baseline gives the conventional McCallum-AvW $\sim 8\times$ home-bias multiplier; the sweep shifts the home coefficient by 3.3 log units in the same direction as the panel PPML.

$d_{\min}$ (km)	$\hat{\gamma}_{\text{OLS}}$			$e^{\hat{\gamma}}$ (OLS log-mean)		
	$\theta = -3$	$\theta = -5$	$\theta = -7$	$\theta = -3$	$\theta = -5$	$\theta = -7$
0.5	-1.155	-2.919	-3.360	$0.32\times$	$0.05\times$	$0.03\times$
1.0	-0.365	-2.054	-2.494	$0.69\times$	$0.13\times$	$0.08\times$
5.0	+1.114	-0.378	-0.819	$3.05\times$	$0.68\times$	$0.44\times$
10	+1.601	+0.209	-0.231	$4.96\times$	$1.23\times$	$0.79\times$
50	+2.336	+1.282	+0.842	$10.3\times$	$3.60\times$	$2.32\times$
100	+2.499	+1.580	+1.140	$12.2\times$	$4.86\times$	$3.13\times$
500	+2.610	+2.031	+1.590	$13.6\times$	$7.62\times$	$4.91\times$
CEPII baseline ($\theta = -1$)		+2.132		$8.43\times$		

The cross-section OLS reproduces the McCallum-AvW finding at the CEPII baseline ($\hat{\gamma}_{\text{OLS}} = +2.13$, multiplier $8.43\times$, in the conventional border-puzzle range), and the same monotonic $d_{\min}$ pattern as the panel PPML, with a sign-flip between $d_{\min} = 1$ and $d_{\min} = 2$ km at $\theta = -3$. The *direction* of the regularization-sensitivity finding is therefore robust to estimator choice; the *magnitude* differs because the two estimators target different conditional means.

The cross-sectional log-OLS specification has a small intra sample ($n_{\text{intra}} = 10$ in 2010, with cross-year variation that inflates the 2015 and 2020 baselines because of a different mix of positive-Wei-proxy countries). The panel PPML preempts this concern by stacking three years against 653 intra rows and absorbing multilateral resistance via origin-year and destination-year FE, which is why I report panel PPML as the primary specification.

I also explored a cross-section PPML with country FE (`statsmodels.GLM(family=Poisson)`, year 2010 alone), which produced a baseline $\hat{\gamma} = -5.01$ with convergence warnings on every cell of the sweep. The high-dimensional FE structure exceeds the BFGS optimizer’s reach in `statsmodels`; the panel PPML uses the dedicated alternating-projection algorithm (Correia et al., 2020) and converges cleanly. The cross-section PPML directional pattern matches both the OLS and panel PPML results.

6 Implications

6.1 The border puzzle as partly a measurement artifact

The McCallum (1995) finding that intra-Canadian trade is roughly $22\times$ larger than equivalent Canadian-US trade, and the Anderson-van Wincoop (2003) “border puzzle” resolution that reduces

the multiplier to roughly $5\times$ via the multilateral resistance term, both depend on a measured intra-national distance. Theorem 1 implies that any measured intra-national distance at $\theta \leq -2$ is mechanically a function of the regularization grain. The empirical result in Section 5 quantifies how much of the remaining multiplier is regularization-driven.

6.2 What this means for applied gravity practice

Two operational implications follow.

First, replacing CEPII's static `distw_harmonic` with a satellite-calibrated time-varying alternative (the contribution of [Gonchar and Helfrich 2026](#)) does not resolve the underlying structural inconsistency. It substitutes one regularization choice for another. The structurally consistent procedure is to acknowledge that d_{ii} is not a well-defined object at gravity-consistent σ and to estimate the model in a way that does not pin d_{ii} to a single number.

Second, the convention of reporting border-puzzle multipliers as structural objects, rather than as conditional on a specific regularization, has misrepresented the literature's empirical content. A border-puzzle multiplier of $10\times$ at $d_{\min} = 100$ km and $5\times$ at $d_{\min} = 10$ km are not contradictory results: they are the same regression conditional on different regularization choices. The literature has been comparing apples to oranges.

6.3 Implications for the broader structural-gravity literature

The structural-gravity literature divides into traditions that construct d_{ii} from country geometry (the CEPII tradition, the Gonchar-Helfrich satellite-distance project, any WorldPop-or-GHS-POP-based intra-national distance) and traditions that avoid the spatial construction entirely. The structural critique in this paper applies to the first class. The second class preempts the issue, each in its own way.

[Eaton and Kortum \(2002\)](#) normalize $d_{ii} = 1$ by assumption, treating the intra-national trade cost as the unit of measurement against which all bilateral trade costs are scaled. In their notation, θ is the Fréchet productivity-shape parameter and is strictly positive ($\theta > \sigma - 1 > 0$), so the symbol overload with the CES integration parameter is mechanical, not structural. The divergence-at- $\theta = -2$ pathology does not arise because there is no spatial integration over country geometry.

[Caliendo and Parro \(2015\)](#) estimate trade elasticities sector by sector via a tetrad estimator (the cross-product of bilateral flows) that cancels distance, language, and contiguity from the identifying variation. Domestic sales are constructed as gross production minus exports, a residual that does not invoke any geometric distance. Sector-by-sector θ estimates from [Caliendo and Parro \(2015\)](#) fall in $[4, 10]$, consistent with the structural range invoked here, but their identification does not pass through d_{ii} .

[Costinot et al. \(2012\)](#) identify θ off cross-industry productivity dispersion, absorbing all bilateral trade costs in importer-exporter fixed effects. Their headline $\hat{\theta} = 6.53$ is in the structural range; their estimation strategy avoids any direct use of intra-national distance.

The three structural-gravity traditions that side-step d_{ii} construction therefore preempt the critique. The critique applies specifically to the Head-Mayer-CEPII tradition, which is the dominant applied-gravity workhorse and the only tradition that constructs the intra-national distance from a CES spatial integral.

6.4 Implications for the Anderson-van Wincoop multilateral-resistance system

The [Anderson and van Wincoop \(2003\)](#) solution to the border puzzle splits the McCallum multiplier into a structural part (the multilateral resistance terms Π_i and P_j) and a residual that they attribute to the border. Their decomposition uses CEPII-style intra-national distance as an input to the MR system; the implied border effect inherits the same regularization sensitivity as the underlying d_{ii} . A re-estimation of the Anderson-van Wincoop system with explicit regularization-grain documentation, and with the structurally consistent $\theta = \delta(1 - \sigma)$ where the integral is divergent, is a natural follow-up. I suspect the Anderson-van Wincoop $5\times$ multiplier is similarly regularization-conditional.

7 Limitations and future work

The paper has six limitations that a careful reader should hold in mind, and four follow-up projects that the diagnosis naturally points to.

7.1 Limitations

First, the empirical demonstration is on the Wei (1996) intra-national flow proxy. The Wei proxy treats GDP minus exports as intra-national trade and is well-known to be noisy: it conflates services, government spending, and non-traded production with intra-national trade in goods. The modern Yotov-Piermartini-Monteiro-Larch (2016) “trade for production” proxy uses sectoral GDP and is the empirical gold standard. The structural critique here applies under either proxy; the magnitudes of the home coefficient may shift modestly under the YPMML proxy. Adapting the regularization sweep to sectoral data is a clean robustness extension and is in scope for a follow-up paper.

Second, the paper’s empirical claim is sensitive to the iceberg-cost exponent δ . I take $\delta = 1$ throughout, which is the standard structural-gravity assumption. If empirical evidence points to $\delta < 1$ (e.g., [Hummels 2007](#) freight-cost meta-evidence suggests freight elasticities of ~ 0.2 - 0.4), then the structurally consistent $\theta = \delta(1 - \sigma)$ can fall in $[-2, 0]$ rather than $[-7, -3]$, above the pole. The structural critique is conditional on $\delta = 1$. Under $\delta < 1$, the divergence is averted and the structural and reduced-form θ can be reconciled with a small change in the iceberg-cost specification. The paper should be read as: *either* the structural critique applies (under $\delta = 1$) *or* the literature must commit to $\delta \ll 1$. Both choices have consequences that the literature has not been explicit about.

Third, the empirical sample is three years. I use 2010, 2015, and 2020 as a triennial cross-section for the panel PPML. A full annual panel (2000-2022) would be standard for a published gravity paper and is the next computational step. I expect the qualitative finding to survive but the standard errors to shrink considerably.

Fourth, the regularization sweep uses a uniform-disk approximation for the country geometry. Real countries have irregular shape, non-uniform population density, and internal heterogeneity. The implied $d_{ii}^{\text{eff}}(\theta, d_{\min})$ on a real country polygon differs from the uniform-disk simulation by country-specific factors. The scaling exponent $\alpha(\theta) = (\theta + 2)/\theta$ is dimension-dependent (2D) but does not depend on the specific shape; the qualitative pattern survives. Quantitative extension to real polygons via the Gonchar-Helfrich satellite-distance panel is the obvious follow-up.

Fifth, the cross-section log-OLS robustness check has small intra samples ($n_{\text{intra}} \in [6, 10]$ across the three years). The panel PPML preempts this concern by stacking years and absorbing multilateral resistance; the log-OLS results are reported as a robustness check, not as the primary empirical foundation.

Sixth, the structural critique is theoretical; the empirical demonstration is one estimator family. I show the home coefficient is regularization-sensitive in log-OLS and in PPML, both as cross-section and panel. The structural critique applies regardless of estimator (the integral diverges as a mathematical fact). A definitive demonstration that the regularization sensitivity propagates into welfare counterfactuals (e.g., Caliendo-Parro 2015 exact-hat methodology) is a natural follow-up.

7.2 Future work

Paper 5b (concept): Regularization-aware structural estimation. Develop an estimator that takes $d_{\min}$ as an explicit input parameter and returns conditional structural estimates. The analyst would then report results as a function of $d_{\min}$, not as a single number. This is the constructive proposal that follows from the diagnosis in Paper 5a.

Paper 5c (concept): Welfare counterfactuals when d_{ii} does not exist as a fixed object. Apply the Caliendo-Parro exact-hat methodology to a regularization-sensitive intra-national distance and quantify how much the implied welfare effects of a trade-policy change depend on the regularization. The structural critique becomes operational at the policy-counterfactual level.

Paper 5d (concept): An honest re-estimation of the Anderson-van Wincoop multilateral-resistance system with explicit regularization-grain documentation and structurally consistent θ . The conjecture: the AvW $5\times$ multiplier shrinks materially when d_{ii} is taken seriously, possibly below $1\times$ at fine spatial resolution.

The Gonchar-Helfrich satellite-distance panel (in flight, target Journal of International Economics) provides the empirical machinery for these extensions. The companion paper documents the satellite-calibrated bilateral distance variants 2000-2024 using WorldPop, VIIRS, and OpenStreetMap; Paper 5a’s structural critique is the methodological scaffolding that makes the satellite-distance project’s reporting honest about its regularization choices.

8 Conclusion

The Head-Mayer (2002) closed-form expression for intra-national distance is a finite analytical object at the parameter values where Head and Mayer evaluated it ($\theta \in [-1.5, 1]$). At the structural CES integration parameter that the gravity literature claims to be estimating ($\theta = \delta(1-\sigma) \in [-7, -3]$ for empirically defensible σ), the underlying integral diverges in 2D. CEPII’s choice of $\theta = -1$ for its `distw` construction is the ϵ -above-1D-divergence value, evaluated through a 2D formula, and is mathematically valid as the harmonic-mean radial coordinate. It is not the structural CES expected pairwise distance.

The gap between these two interpretations of θ is the structural-versus-reduced-form conflation. Head and Mayer (2002) performed it themselves, citing “hundreds of gravity equation estimates of θ ” as evidence for the parameter choice. Mayer and Zignago (2011) propagated it into the CEPII database. Head and Mayer (2014) retained the symbol overload as a deliberate choice, “to emphasize the similarity in resulting terms.” The applied gravity literature has built on this convention.

The consequences are quantifiable. The home-bias coefficient in a gravity regression depends on the spatial resolution at which intra-national distance is constructed, and the dependence is steep at structurally consistent θ . A material share of the conventional border-puzzle multiplier is a measurement artifact. The structurally-correct response is not to compute d_{ii} better, but to acknowledge that the integral does not exist as a fixed property of the geometry at gravity-consistent σ , and to design empirical specifications that do not require pinning it to a single number.

A Notation

A glossary of symbols used in the paper. Reduced-form θ refers to the regression coefficient of $\log X_{ij}$ on $\log d_{ij}$. Structural θ refers to the CES integration parameter $\delta(1 - \sigma)$. The two are equal only by coincidence.

B Numerical verification of Table 1

Computed in Python via direct substitution into equation (1), using `numpy.power` on `complex128` arrays for $\theta \leq -2$.

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